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Inventor(s)

CASEY; Tamara Lynne et al.

SYSTEMS AND METHODS FOR MANAGING RADIO FREQUENCY SPECTRUM IN GROUND TO AERIAL VEHICLE COMMUNICATIONS

Abstract

Disclosed herein is a terrestrial to air communications network that can be configured to include a spectrum management system that deterministically allocates spectrum to aircraft for use during flight. In one or more examples, a user transmits a flight plan to a spectrum management system that is configured to manage the RF spectrum in a given air space. In one or more examples, and based on the received flight plan, the spectrum management system can allocate an RF spectrum frequency “slot” (i.e., timeslot, subchannel, or resource block) for the aircraft to use during its intended flight. The spectrum management system can take into account available spectrum as well as the predicted network traffic and their spectrum allocations to determine an RF spectrum slot that can provide a stable and continuous communications channel to the aircraft during its flight.

Inventors: CASEY; Tamara Lynne (Annapolis, MD), GAGNE; Michael Robert (Monrovia, MD)

Applicant: AURA Network Systems, Inc. (McLean, VA)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/608,692, filed Mar. 18, 2024, and published on Jul. 4, 2024 as U.S. Publication No. 2024-0224309, which is a continuation of U.S. patent application Ser. No. 17/569,379, filed Jan. 5, 2022, and issued on Apr. 16, 2024 as U.S. Pat. No. 11,963,203, which claims benefit to U.S. Provisional Application No. 63/293,579, filed Dec. 23, 2021, U.S. Provisional Application No. 63/217,466, filed Jul. 1, 2021, and U.S. Provisional Application No. 63/134,521, filed Jan. 6, 2021, the entire contents of each of which are incorporated herein by reference.

FIELD

[0002] This disclosure relates to RF spectrum management in an air to ground communications network, and more specifically, to systems and methods for allocating RF narrowband spectrum channels to airborne assets to facilitate communications with a ground-based communications network.

BACKGROUND

[0003] One of the critical features of air safety for both manned and unmanned flights is the ability for airborne assets to communicate with the ground so as to relay operationally critical communications. Ensuring that an airborne asset can maintain a continuous and uninterrupted communications link to the ground ensures that the airborne asset is able to receive necessary information from ground controllers as well as transmit necessary information to the ground controllers at any and all points during a given flight.

[0004] The proliferation of airborne assets, and specifically unmanned aerial vehicles (UAVs) has complicated the task of ensuring that each airborne asset has continuous communications channel with a ground station. UAVs are now capable of flying long distances across a broad array of geographic areas, all the while having specialized communications requirements with the ground. For instance, UAV operators based on the ground must be in constant communication with the UAV not only to provide instructions to the UAV from the ground, but to also receive critical telemetry from the UAV that informs the ground-based operator about the UAV's operational status.

[0005] Ensuring the performance of the critical data link between a ground base station and remote radios for aviation operations in airspace is critical to supporting the safety requirements of manned, unmanned and optionally piloted flights. Data links need to meet the reliability, integrity and availability performance targets set forth by regulators. Ensuring a continuous data link for remote radios can be challenging in environment in which there are many airborne assets are transiting an airspace at any given time. The availability of RF spectrum specifically can be a

challenging issue. With increasing air traffic comes an increasing potential for the communications of one aircraft to interfere with the communications of another during a flight. Network interference can further be exacerbated when aircraft in a given network are transiting across large geographic areas and thus must rely on multiple ground base stations during its flight to maintain a continuous communications link with the ground.

SUMMARY

[0006] According to an aspect, a terrestrial to air communications network can be configured to include a spectrum management system that deterministically allocates spectrum to aircraft for use during flight. In one or more examples, a user wishing to fly an aircraft transmits a flight plan to a spectrum management system that is configured to manage the RF spectrum in a given air space. In one or more examples, the flight plan can include the intended geographic route of the aircraft, the beginning time of the flight, the expected end time of the flight, as well as operational details of the flight such as the data throughput requirement and the radio configuration of the aircraft. In one or more examples, and based on the received flight plan, the spectrum management system can allocate an RF spectrum frequency “slot” (i.e., timeslot, subchannel, or resource block) for the aircraft to use during its intended flight. In one or more examples, the spectrum management system can choose what spectrum slot to give an aircraft based not only on its flight plan but on other various factors such as a dynamic link budget created for the flight, dynamic RF coverage predictions, and dynamic interference and co-existence (i.e., with other aircraft in time, space, and RF channel utilization) predictions. The spectrum management system can take into account available spectrum as well as the predicted network traffic and their spectrum allocations to determine an RF spectrum slot that can provide a stable and continuous communications channel to the aircraft during its flight.

[0007] According to an aspect, the flight plan based dynamic spectrum/traffic channel management system (i.e., the spectrum management system) can reserve and assign spectrum resources available at ground base stations, remote ground based radios and/or remote airborne radios. In one or more examples these spectrum resources can be in both the frequency and time domain of a digital system and include spectrum bandwidth and individual traffic channels called sub-channels or resource blocks within the spectrum bandwidth. In one or more examples, a traffic channel (also referred throughout the disclosure as a sub-channels, frames, resource block, and bearers) can refer to a communications link at a specified frequency that allows data communication between an aircraft and a pilot on an aviation network. In one or more examples, a traffic channel can be represented as a plurality of sub-channels, frames, resource blocks, and bearers. The system looks at several variables such as the available spectrum resources, radio link throughput and performance requirements, location (including elevation), time period as well as the radio frequency environment to assign a non-contended resource between the ground based radios and remote radios. In one or more examples, these variables can be populated by several internal and external components to the spectrum management system. Due to the large volume of information and the potential for spectrum and/or traffic channel requests by thousands of end users in the spectrum management system can include a “digital twin” of the spectrum management system that can be configured to perform the required analysis without affecting the operational system.

[0008] According to an aspect, once the spectrum management system determines the RF spectrum slot to provide to an aircraft, the system can transmit the selected RF spectrum slot and additional information about the RF channel (such as modulation scheme, error correction code, etc.) to one or more base stations in the communications network that take the information and execute the RF communications channel during the flight. In one or more examples, the ground base station can include a spectrum monitoring system that is configured to continuously monitor the active RF environment of a given airspace for interference. In one or more examples, if undesired interference is detected that has the potential to impact flight operations, the ground base station can utilize an antenna with null forming, beam steering technology that can direct an interference

signal (i.e., a “null”) at the unwanted interference source to minimize or remove the interference. In one or more examples, rather than employing a ground-based beam forming antenna, the system can utilize a beam forming antenna that is employed on a satellite.

[0009] According to an aspect, a method for allocating RF spectrum channels in an air-to-ground communications network includes receiving a flight plan from a user, wherein the flight plan comprises timing, location, and altitude information for a flight that is to fly in one or more coverage areas of the air-to-ground communications network, determining RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user, selecting one or more RF spectrum channels from a traffic channel pool based on the received flight plan, determining the presence of one or more interfering signals in the communications network based on the received flight plan the selected one or more RF spectrum channels from the traffic channel pool, and if it is determined that there are no interfering signals in the communications networks, reserving the selected one or more RF spectrum channels for the received flight plan.

[0010] Optionally, the flight plan comprises information about the radio configuration of an aircraft performing the flight plan.

[0011] Optionally, the method comprises determining a throughput requirement of the aircraft based on the received information about the radio configuration of the aircraft performing the flight plan.

[0012] Optionally, the flight plan comprises a throughput requirement of an aircraft performing the flight plan.

[0013] Optionally, determining RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user comprises: generating one or more dynamic link budgets for the flight based on the received flight plan, wherein each dynamic link budget is configured to determine RF availability of a communications link in one or more coverage areas of the communications network, generating one or more geofences based on the dynamic link budget, and determining RF availability based on the dynamic link budgets corresponding to the one or more coverage areas of the communications network that the aircraft is to fly in based on the received flight plan from the user.

[0014] Optionally, generating the one or more dynamic link budgets for the flight includes using a dynamic RF coverage prediction tool to populate one or more parameters of the one or more dynamic link budgets.

[0015] Optionally, selecting one or more RF spectrum channels from the traffic channel pool comprises determining a number of RF spectrum channels to assign to the flight based on a throughput requirement of the flight.

[0016] Optionally, selecting one or more RF spectrum channels from the traffic channel pool comprises selecting the determined number of RF spectrum channels from the available RF spectrum channels in the traffic channel pool.

[0017] Optionally, the traffic channel pool comprises one or more reserved RF spectrum channels, wherein the reserved RF spectrum channels comprise RF spectrum channels that have been already reserved by another flight of the communications network.

[0018] Optionally, the traffic channel pool comprises one or more restricted traffic channels, wherein the restricted RF spectrum channels comprise RF spectrum channels reserved for use by the spectrum management system.

[0019] Optionally, determining the presence of one or more interfering signals in the communications network comprises using a dynamic interference prediction tool.

[0020] Optionally, the method comprises transmitting the selected one or more RF spectrum channels to a third-party regulatory body for approval, and reserving the selected one or more RF spectrum channels for the received flight plan upon receiving approval from the regulatory body.

[0021] Optionally, the method comprises transmitting the selected one or more RF spectrum

channels to a spectrum management system.

[0022] Optionally, the base station controller is configured to operate a communications link between a pilot and an aircraft used to execute the received flight plan during the flight.

[0023] Optionally, the method comprises selecting a modulation scheme to operate the communications link between the pilot and the aircraft.

[0024] Optionally, the method comprises selecting a forward error correction code to operate the communications link between the pilot and the aircraft.

[0025] According to an aspect, A method for implementing and maintaining an RF communications link in an air-to-ground communications network includes receiving information about a flight to take place in one or more coverage areas of the air-to-ground communications network, wherein the information comprises one or more RF spectrum channels associated with the flight, generating an RF communications link at a base station of the communications network between a user and an aircraft flying the flight based on the one or more RF spectrum channels associated with the flight, generating a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network, determining the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget, if one or more performance degradations are determined to be present in the RF communications link, applying one or more mitigations to the RF communications link, and updating the dynamic link budget based on the applied one or more mitigations.

[0026] Optionally, the information about a flight to take place in one or more coverage areas of the air-to-ground communications network comprises timing, location, and altitude information for the flight that is to fly in the one or more coverage areas of the air-to-ground communications network.

[0027] Optionally, the information comprises information about the radio configuration of the aircraft performing the flight plan.

[0028] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises transmitting data from the user to the aircraft at an RF frequency based on the one or more RF spectrum channels associated with the flight.

[0029] Optionally, wherein generating an RF communications link at the base station between the user and the aircraft comprises selecting a modulation scheme for communications between the user and the aircraft.

[0030] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises selecting a forward error correction code for communications between the user and the aircraft.

[0031] Optionally, generating a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network comprises receiving information from a base station associated with one or more RF environmental conditions at the base station.

[0032] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base station is generated by a spectrum monitoring device located at the base station.

[0033] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base stations is generated by a base station link monitoring tool configured to detect one or more conditions of the RF communications link between the user and the aircraft.

[0034] Optionally, the dynamic link budget is based on the information generated by the spectrum monitoring device and the base station link monitoring tool.

[0035] Optionally, the spectrum monitoring device comprises a software defined receiver configured to receive and process RF signals received at a monitoring antenna of the base station.

[0036] Optionally, the spectrum monitoring device comprises noise floor monitoring tool configured to detect a power level of a noise floor at the base station.

[0037] Optionally, the spectrum monitoring device comprises a undesired signal detection component configured to detect RF signals that are not expected at the base station.

[0038] Optionally, the spectrum monitoring device comprises a direction and power detection component configured detect a power and direction of the RF signals that are not expected at the base station.

[0039] Optionally, applying one or more mitigations to the RF communications link comprises adjusting one or more parameters associated with the dynamic link budget.

[0040] Optionally, adjusting one or more parameters associated with the dynamic link budget comprises increasing a power of the signal transmitted by an antenna of the base station to the aircraft.

[0041] Optionally, determining the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget comprises determining the presence of one or more interference signals in the RF communications link.

[0042] Optionally, determining the presence of one or more interference signals in the RF communications link comprises determining a direction at the base station from which the one or more interference signals are propagating.

[0043] Optionally, applying one or more mitigations to the RF communications link comprises controlling a beam steering antenna to transmit a null signal in the direction from which the one or more interference signals are propagating.

[0044] According to an aspect, A system for allocating RF spectrum channels in an air-to-ground communications network includes a memory, one or more processors, wherein the memory stores one or more programs that when executed by the one or more processors, cause the one or more processors to receive a flight plan from a user, wherein the flight plan comprises timing, location, and altitude information for a flight that is to fly in one or more coverage areas of the air-to-ground communications network, determine RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user, select one or more RF spectrum channels from a traffic channel pool based on the received flight plan, determine the presence of one or more interfering signals in the communications network based on the received flight plan the selected one or more RF spectrum channels from the traffic channel pool, and if it is determined that there are no interfering signals in the communications networks, reserve the selected one or more RF spectrum channels for the received flight plan.

[0045] Optionally, the flight plan comprises information about the radio configuration of an aircraft performing the flight plan.

[0046] Optionally, the one or more processors are caused to determine a throughput requirement of the aircraft based on the received information about the radio configuration of the aircraft performing the flight plan.

[0047] Optionally, the flight plan comprises a throughput requirement of an aircraft performing the flight plan.

[0048] Optionally, determining RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user comprises generating one or more dynamic link budgets for the flight based on the received flight plan, wherein each dynamic link budget is configured to determine RF availability of a communications link in one or more coverage areas of the communications network, generating one or more geofences based on the dynamic link budget, and determining RF availability based on the dynamic link budgets corresponding to the one or more coverage areas of the communications network that the aircraft is to fly in based on the received flight plan from the user.

[0049] Optionally, generating the one or more dynamic link budgets for the flight includes using a dynamic RF coverage prediction tool to populate one or more parameters of the one or more dynamic link budgets.

[0050] Optionally, selecting one or more RF spectrum channels from the traffic channel pool

comprises determining a number of RF spectrum channels to assign to the flight based on a throughput requirement of the flight.

[0051] Optionally, selecting one or more RF spectrum channels from the traffic channel pool comprises selecting the determined number of RF spectrum channels from the available RF spectrum channels in the traffic channel pool.

[0052] Optionally, the traffic channel pool comprises one or more reserved RF spectrum channels, wherein the reserved RF spectrum channels comprise RF spectrum channels that have been already reserved by another of the communications network.

[0053] Optionally, the traffic channel pool comprises one or more restricted traffic channels, wherein the restricted RF spectrum channels comprise RF spectrum channels reserved for use.

[0054] Optionally, determining the presence of one or more interfering signals in the communications network comprises using a dynamic interference prediction tool.

[0055] Optionally, the one or more processors are caused to transmit the selected one or more RF spectrum channels to a third-party regulatory body for approval, and reserve the selected one or more RF spectrum channels for the received flight plan upon receiving approval from the regulatory body.

[0056] Optionally, the one or more processors are caused to transmit the selected one or more RF spectrum channels to a spectrum management system.

[0057] Optionally, a base station controller is configured to operate a communications link between a pilot and an aircraft used to execute the received flight plan during the flight.

[0058] Optionally, the one or more processors are caused select a modulation scheme to operate the communications link between the pilot and the aircraft.

[0059] Optionally, the one or more processors are caused to select a forward error correction code to operate the communications link between the pilot and the aircraft.

[0060] According to an aspect, a system for implementing and maintaining an RF communications link in an air-to-ground communications network includes a memory, one or more processors, wherein the memory stores one or more programs that when executed by the one or more processors, cause the one or more processors to receive information about a flight to take place in one or more coverage areas of the air-to-ground communications network, wherein the information comprises one or more RF spectrum channels associated with the flight, generate an RF communications link at a base station of the communications network between a user and an aircraft flying the flight based on the one or more RF spectrum channels associated with the flight, generate a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network, determine the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget, if one or more performance degradations are determined to be present in the RF communications link, apply one or more mitigations to the RF communications link, and update the dynamic link budget based on the applied one or more mitigations.

[0061] Optionally, the information about a flight to take place in one or more coverage areas of the air-to-ground communications network comprises timing, location, and altitude information for the flight that is to fly in the one or more coverage areas of the air-to-ground communications network.

[0062] Optionally, the information comprises information about the radio configuration of the aircraft performing the flight plan.

[0063] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises transmitting data from the user to the aircraft at an RF frequency based on the one or more RF spectrum channels associated with the flight.

[0064] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises selecting a modulation scheme for communications between the user and the aircraft.

[0065] Optionally, generating an RF communications link at the base station between the user and

the aircraft comprises selecting a forward error correction code for communications between the user and the aircraft.

[0066] Optionally, generating a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network comprises receiving information from a base station associated with one or more RF environmental conditions at the base station.

[0067] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base station is generated by a spectrum monitoring device located at the base station.

[0068] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base stations is generated by a base station link monitoring tool configured to detect one or more conditions of the RF communications link between the user and the aircraft.

[0069] Optionally, the dynamic link budget is based on the information generated by the spectrum monitoring device and the base station link monitoring tool.

[0070] Optionally, the spectrum monitoring device comprises a software defined receiver configured to receive and process RF signals received at a monitoring antenna of the base station.

[0071] Optionally, the spectrum monitoring device comprises a noise floor monitoring tool configured to detect a power level of a noise floor at the base station.

[0072] Optionally, the spectrum monitoring device comprises a undesired signal detection component configured to detect RF signals that are not expected at the base station.

[0073] Optionally, the spectrum monitoring device comprises a direction and power detection component configured detect a power and direction of the RF signals that are not expected at the base station.

[0074] Optionally, applying one or more mitigations to the RF communications link comprises adjusting one or more parameters associated with the dynamic link budget.

[0075] Optionally, adjusting one or more parameters associated with the dynamic link budget comprises increasing a power of the signal transmitted by an antenna of the base station to the aircraft.

[0076] Optionally, determining the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget comprises determining the presence of one or more interference signals in the RF communications link.

[0077] Optionally, determining the presence of one or more interference signals in the RF communications link comprises determining a direction at the base station from which the one or more interference signals are propagating.

[0078] Optionally, applying one or more mitigations to the RF communications link comprises controlling a beam steering antenna to transmit a null signal in the direction from which the one or more interference signals are propagating.

[0079] According to an aspect, A non-transitory computer readable storage medium storing one or more programs for allocating RF spectrum channels in an air-to-ground communications network, for execution by one or more processors of an electronic device that when executed by the device, cause the device to receive a flight plan from a user, wherein the flight plan comprises timing, location, and altitude information for a flight that is to fly in one or more coverage areas of the air-to-ground communications network, determine RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user, select one or more RF spectrum channels from a traffic channel pool based on the received flight plan, determine the presence of one or more interfering signals in the communications network based on the received flight plan the selected one or more RF spectrum channels from the traffic channel pool, and if it is determined that there are no interfering signals in the communications networks, reserve the selected one or more RF spectrum channels for the received flight plan.

[0080] Optionally, wherein the flight plan comprises information about the radio configuration of an aircraft performing the flight plan.

[0081] Optionally, the device is caused to determine a throughput requirement of the aircraft based on the received information about the radio configuration of the aircraft performing the flight plan.

[0082] Optionally, the flight plan comprises a throughput requirement of an aircraft performing the flight plan.

[0083] Optionally, determining RF availability in the one or more coverage areas of the communications network based on the received flight plan from the user comprises generating one or more dynamic link budgets for the flight based on the received flight plan, wherein each dynamic link budget is configured to determine RF availability of a communications link in one or more coverage areas of the communications network, generating one or more geofences based on the dynamic link budget, and determining RF availability based on the dynamic link budgets corresponding to the one or more coverage areas of the communications network that the aircraft is to fly in based on the received flight plan from the user.

[0084] Optionally, generating the one or more dynamic link budgets for the flight includes using a dynamic RF coverage prediction tool to populate one or more parameters of the one or more dynamic link budgets.

[0085] Optionally, selecting one or more RF spectrum channels from the traffic channel pool comprises determining a number of RF spectrum channels to assign to the flight based on a throughput requirement of the flight.

[0086] Optionally, selecting one or more RF spectrum channels from the traffic channel pool comprises selecting the determined number of RF spectrum channels from the available RF spectrum channels in the traffic channel pool.

[0087] Optionally, wherein the traffic channel pool comprises one or more reserved RF spectrum channels, wherein the reserved RF spectrum channels comprise RF spectrum channels that have been already reserved.

[0088] Optionally, the traffic channel pool comprises one or more restricted traffic channels, wherein the restricted RF spectrum channels comprise RF spectrum channels reserved for use.

[0089] Optionally, determining the presence of one or more interfering signals in the communications network comprises using a dynamic interference prediction tool.

[0090] Optionally, wherein the one or more processors are caused to transmit the selected one or more RF spectrum channels to a third-party regulatory body for approval, and reserve the selected one or more RF spectrum channels for the received flight plan upon receiving approval from the regulatory body.

[0091] Optionally, wherein the device is caused to transmit the selected one or more RF spectrum channels to a spectrum management system.

[0092] Optionally, wherein a base station controller is configured to operate a communications link between a pilot and an aircraft used to execute the received flight plan during the flight.

[0093] Optionally, the device is caused to select a modulation scheme to operate the communications link between the pilot and the aircraft.

[0094] Optionally, the device is caused to select a forward error correction code to operate the communications link between the pilot and the aircraft.

[0095] Optionally, A non-transitory computer readable storage medium storing one or more for implementing and maintaining an RF communications link in an air-to-ground communications network for allocating RF spectrum channels in an air-to-ground communications network, for execution by one or more processors of an electronic device that when executed by the device, cause the device to receive information about a flight to take place in one or more coverage areas of the air-to-ground communications network, wherein the information comprises one or more RF spectrum channels associated with the flight, generate an RF communications link at a base station of the communications network between a user and an aircraft flying the flight based on the one or

more RF spectrum channels associated with the flight, generate a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network, determine the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget, if one or more performance degradations are determined to be present in the RF communications link, apply one or more mitigations to the RF communications link, and update the dynamic link budget based on the applied one or more mitigations.

[0096] Optionally, the information about a flight to take place in one or more coverage areas of the air-to-ground communications network comprises timing, location, and altitude information for the flight that is to fly in the one or more coverage areas of the air-to-ground communications network.

[0097] Optionally, the information comprises information about the radio configuration of the aircraft performing the flight plan.

[0098] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises transmitting data from the user to the aircraft at an RF frequency based on the one or more RF spectrum channels associated with the flight.

[0099] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises selecting a modulation scheme for communications between the user and the aircraft.

[0100] Optionally, generating an RF communications link at the base station between the user and the aircraft comprises selecting a forward error correction code for communications between the user and the aircraft.

[0101] Optionally, generating a dynamic link budget for the flight based on the received information and based on one or more conditions of the communications network comprises receiving information from a base station associated with one or more RF environmental conditions at the base station.

[0102] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base station is generated by a spectrum monitoring device located at the base station.

[0103] Optionally, the information from the base station associated with the one or more RF environmental conditions at the base stations is generated by a base station link monitoring tool configured to detect one or more conditions of the RF communications link between the user and the aircraft.

[0104] Optionally, the dynamic link budget is based on the information generated by the spectrum monitoring device and the base station link monitoring tool.

[0105] Optionally, the spectrum monitoring device comprises a software defined receiver configured to receive and process RF signals received at a monitoring antenna of the base station.

[0106] Optionally, the spectrum monitoring device comprises noise floor monitoring tool configured to detect a power level of a noise floor at the base station.

[0107] Optionally, the spectrum monitoring device comprises a undesired signal detection component configured to detect RF signals that are not expected at the base station.

[0108] Optionally, the spectrum monitoring device comprises a direction and power detection component configured detect a power and direction of the RF signals that are not expected at the base station.

[0109] Optionally, applying one or more mitigations to the RF communications link comprises adjusting one or more parameters associated with the dynamic link budget.

[0110] Optionally, adjusting one or more parameters associated with the dynamic link budget comprises increasing a power of the signal transmitted by an antenna of the base station to the aircraft.

[0111] Optionally, determining the presence of one or more performance degradations in the RF communications link during the flight based on the generated dynamic link budget comprises

determining the presence of one or more interference signals in the RF communications link.

[0112] Optionally, determining the presence of one or more interference signals in the RF communications link comprises determining a direction at the base station from which the one or more interference signals are propagating.

[0113] Optionally, applying one or more mitigations to the RF communications link comprises controlling a beam steering antenna to transmit a null signal in the direction from which the one or more interference signals are propagating.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0114] The invention will now be described, by way of example only, with reference to the accompanying drawings, in which:

[0115] FIG. 1 illustrates a ground to air network according to examples of the disclosure.

[0116] FIG. 2 illustrates an exemplary system for RF spectrum management of ground to air communication networks according to examples of the disclosure.

[0117] FIG. 3 illustrates an exemplary system for RF spectrum allocation and management according to examples of the disclosure.

[0118] FIG. 4A illustrates an exemplary traffic channel pool according to examples of the disclosure.

[0119] FIG. 4B illustrates another exemplary traffic channel pool according to examples of the disclosure.

[0120] FIG. 5 illustrates an exemplary channel allocation process for RF spectrum management of ground to air communication networks according to examples of the disclosure.

[0121] FIG. 6 illustrates an exemplary spectrum monitoring device according examples of the disclosure.

[0122] FIG. 7 illustrates an exemplary beam steering antenna system according to examples of the disclosure.

[0123] FIG. 8 illustrates an exemplary in-flight process for RF spectrum management of ground to air communication networks according to example of the disclosure.

[0124] FIG. 9 illustrates an exemplary system for network of networks according to examples of the disclosure.

[0125] FIG. 10 illustrates an exemplary computing system, according to examples of the disclosure.

DETAILED DESCRIPTION

[0126] Reference will now be made in detail to implementations and embodiments of various aspects and variations of systems and methods described herein. Although several exemplary variations of the systems and methods are described herein, other variations of the systems and methods may include aspects of the systems and methods described herein combined in any suitable manner having combinations of all or some of the aspects described.

[0127] Described herein are systems and methods for assigning and managing RF communications between ground-based stations and airborne assets. In one or more examples of the disclosure, a pilot or other user can generate and transmit a flight plan to a spectrum management system. Additionally, or alternatively, the pilot or use can also transmit additional information to the spectrum management system such as the type of aircraft/radio configuration that they will be using during a flight, and a request for a certain amount of data throughput that they want to have access to during the flight.

[0128] In one or more examples, upon receiving the flight plan and/or information from the pilot, the spectrum management system can proceed to match the user with an RF spectrum slot for the

pilot to use during their planned flight. As described in further detail below, the selection of the slot for the pilot's use by the spectrum management system can be based on the information and flight plan submitted by the pilot, as well as other factors such as predictions about RF spectrum interference that may occur during the flight, RF coverage prediction, and RF spectrum availability in the geographic area or areas that the aircraft will traverse during their flight.

[0129] According to various embodiments, once the RF spectrum slot or slots has been selected for a given flight plan, the spectrum management system can also be configured to implement a communications link between the pilot/operator and the aircraft using the RF spectrum slot assigned to the aircraft and its flight plan. In one or more examples, the spectrum management system can configure the communications link between an operator and a plane by applying the chosen RF spectrum slot to a ground base station and selecting other parameters associated with the communications link.

[0130] In the following description of the various embodiments, it is to be understood that the singular forms “a,” “an,” and “the” used in the following description are intended to include the plural forms as well, unless the context clearly indicates otherwise. It is also to be understood that the term “and/or” as used herein refers to and encompasses any and all possible combinations of one or more of the associated listed items. It is further to be understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used herein, specify the presence of stated features, integers, steps, operations, elements, components, and/or units but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, units, and/or groups thereof.

[0131] Certain aspects of the present disclosure include process steps and instructions described herein in the form of an algorithm. It should be noted that the process steps and instructions of the present disclosure could be embodied in software, firmware, or hardware and, when embodied in software, could be downloaded to reside on and be operated from different platforms used by a variety of operating systems. Unless specifically stated otherwise as apparent from the following discussion, it is appreciated that, throughout the description, discussions utilizing terms such as “processing,” “computing,” “calculating,” “determining,” “displaying,” “generating” or the like, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system memories or registers or other such information storage, transmission, or display devices.

[0132] The present disclosure in some embodiments also relates to a device for performing the operations herein. This device may be specially constructed for the required purposes, or it may comprise a general purpose computer selectively activated or reconfigured by a computer program stored in the computer. Such a computer program may be stored in a non-transitory, computer readable storage medium, such as, but not limited to, any type of disk, including floppy disks, USB flash drives, external hard drives, optical disks, CD-ROMs, magnetic-optical disks, read-only memories (ROMs), random access memories (RAMs), EPROMs, EEPROMs, magnetic or optical cards, application specific integrated circuits (ASICs), or any type of media suitable for storing electronic instructions, and each connected to a computer system bus. Furthermore, the computing systems referred to in the specification may include a single processor or may be architectures employing multiple processor designs, such as for performing different functions or for increased computing capability. Suitable processors include central processing units (CPUs), graphical processing units (GPUs), field programmable gate arrays (FPGAs), and ASICs.

[0133] The methods, devices, and systems described herein are not inherently related to any particular computer or other apparatus. Various general-purpose systems may also be used with programs in accordance with the teachings herein, or it may prove convenient to construct a more specialized apparatus to perform the required method steps. The required structure for a variety of these systems will appear from the description below. In addition, the present invention is not described with reference to any particular programming language. It will be appreciated that a

variety of programming languages may be used to implement the teachings of the present disclosure as described herein.

[0134] When an aircraft is in-flight, it is critical that the aircraft have a reliable and continuous communication link with the ground. For instance, in the context of unmanned aircraft vehicles (UAVs) in which an aircraft is flown and controlled from the ground by an operator, the operator will need to be continuously updated with information regarding the status of the UAV. In order to facilitate unmanned flight, UAVs must have continuous contact with operators on the ground so that they can receive instructions and also so that they can transmit vital telemetry information to let the operators know the status of the flight. However, as air traffic increases across the world, providing a reliable and continuous communications link to an aircraft for the entire duration of its flight can become a complicated endeavor. A ground-to-air communications network can include many aircraft, ground stations, and geographic areas that need to be coordinated so as to ensure that a single aircraft in the network can be provided with a reliable and continuous communications channel during its flight.

[0135] FIG. 1 illustrates a ground to air network according to examples of the disclosure. The example of FIG. 1 illustrates an exemplary communications network **100** that can be configured to provide communications between one or more ground base stations **104a-c** and one or more aircraft **102** in-flight. In one or more examples, the communications network **100** can include one or more ground base stations **104a-c**. Each of the ground base stations **104a-c** include one or more antennas configured to transmit communications from the ground to the one or more aircraft **102**. In one or more examples, each ground base station **104a-c** can be configured to provide transmissions within a coverage area **108a-c**. For example, ground base station **104a** can be configured to transmit RF spectrum radio signals over geographic coverage area **108a**. Ground base station **104b** can be configured to transmit RF spectrum radio signals over geographic coverage area **108b**, and ground base station **104c** can be configured to transmit RF spectrum radio signals over geographic coverage area **108c**. In one or more examples, geographic coverage areas **108a-c** can be three-dimensional areas that not only cover a certain range of latitude and longitude, but also provide coverage to areas from the ground up until a maximum serviceable altitude.

[0136] In one or more examples, each aircraft **102** can be handed over from one ground base station to the next during the duration of its flight. For instance, at the beginning of a flight, ground base station **104a** can be responsible for providing a communications channel between an operator on the ground and the aircraft while the aircraft **102** is within the coverage area **108a**. If during the flight, the aircraft transits out of the coverage area **108a** into coverage area **108b**, then responsibility for providing the communications channel can transition from ground base station **104a** to ground station **104b**. If during the flight, the aircraft **102** transits out of coverage area **108b** into coverage area **108c**, then responsibility for providing the communications channel can transition from ground base station **104b** to ground station **104c**. In this way, the communications network **100** can be configured to ensure that an aircraft has an established communications channel with at least one ground base station at any point along its flight plan, so long as the flight plan passes through at least one coverage area at any point during its flight.

[0137] In one or more examples, each base station **104a-c** can be communicatively coupled to a base station controller **106a-c** respectively. Thus, in one or more examples, ground base station **104a** can be communicatively coupled to base station controller **106a**, ground base station **104b** can be communicatively coupled to base station controller **106b**, and ground base station **104c** can be communicatively coupled to base station controller **106c**. As described in further detail below, each base station controller can be configured to implement an RF based communications channel between a ground operator and an aircraft **102** when the aircraft is transiting through the coverage area **108a-c** that corresponds to the base station that the controller is configured to operate. In one or more examples, implementing an RF based communications channel can include modulating signals transmitted by the operator to a RF spectrum frequency assigned to the aircraft **102**,

applying an appropriate modulation scheme to the transmitted signals, and applying any other physical layer communications protocols such as error correction codes.

[0138] In one or more examples, the goal of the communications network **100** can be to provide any given aircraft **102** operating within the network with a continuous and reliable RF spectrum channel throughout the duration of its flight. In one or more examples, providing a continuous and reliable RF spectrum to an aircraft can include providing a single RF spectrum channel (i.e., slot) to an aircraft that it can reliably use throughout the duration of its flight to communicate with the ground. In one or more examples, each aircraft in a given airspace can communicate with the ground using a dedicated RF spectrum channel (i.e., a frequency range in the RF spectrum that is unique to the aircraft and can be only used by that individual aircraft to transmit and receive communications from the ground). In order to facilitate efficient flight operations, in one or more examples, each ground base station **104a-c** coupled to its corresponding base station controller **106a-c** can be configured to ensure that each aircraft in its coverage area **108a-c** is able to communicate with the ground using communications transmitted in the RF spectrum channel assigned to that aircraft.

[0139] Assigning aircraft a dedicated RF spectrum channel to use throughout the duration of its flight can be difficult. Often times a given base station is responsible for providing communications channels to hundreds of flights at any given moment in time, with each aircraft in the coverage needing its own dedicated RF spectrum channel so that it can communicate with the ground uninterrupted by other air traffic in the air space. Furthermore, since flights can traverse multiple base stations during a given flight, assigning a dedicated RF spectrum to a flight that can be used throughout the flights duration to have non-contended communications with the ground can require a high level of coordination to ensure that no two aircraft transit the same coverage area using the same RF spectrum channel. Furthermore, as the RF environment in a given coverage area is dynamic, not only is there a need to make sure that no two flights in a given coverage area operate on the same RF channel, but there is also a need to make sure that any communications between an aircraft and the ground will not be interfered with from various noise sources that can operate in a given coverage area. These noise sources can include the RF Noise Floor, related or unrelated network Co-Channel or Adjacent Channel Interference as well as Out of Band Interference sources.

[0140] In order to coordinate the assignment of RF spectrum channels to aircraft, in one or more examples, a system for Dynamic Spectrum Management that is configured to support safe aviation operations can be implemented to coordinate RF spectrum channel allocation to aircraft operating in a given communications network. In one or more examples, the spectrum management system can allocate the spectrum and traffic channels in a deterministic way to ensure the radio resources are available between the ground base station and the airborne radios operating on aircraft in the network.

[0141] FIG. 2 illustrates an exemplary system for RF spectrum management of ground to air communication networks according to examples of the disclosure. In one or more examples of the disclosure, the communications network **200** of FIG. 2 can include the same components (i.e., aircraft **102**, ground base stations **104a-c**, and base station controller **106a-c**) as the communications network **100** described above with respect to FIG. 1, but can also include one or more spectrum management system components (described in further detail below) that can manage the process of allocating RF spectrum channels to aircraft **102** in the network **200**.

[0142] In one or more examples of the disclosure, one or more pilots/operators **206** can be connected to the network **200** in order to transmit data (such as command and control data) to the one or more aircraft. Each of the pilots **206** can be communicatively coupled to the network **200** through a spectrum management system **202** that can be configured to allocate RF spectrum channels to each of the aircraft **102** being controlled by the pilots **206**. In one or more examples, the spectrum management system **202** can be configured to facilitate a communications link between each pilot **206** and their corresponding aircraft **102** by establishing an RF communications link

using a specified RF spectrum channel allocated to each aircraft.

[0143] In one or more examples of the disclosure, the spectrum management system can be configured to manage each communication link between an aircraft **102** and a pilot/operator **206** in real-time. Thus, in one or more examples, if the spectrum management system **202** determines that a given communications link has been compromised or had degraded, the spectrum management system **202** can take action to adjust the communications link to mitigate the issue. For instance, in one or more examples, if a given RF spectrum channel being used by an aircraft **102** is no longer performing satisfactorily or to required specifications, the spectrum channel management system **202** can change the RF spectrum channel (described in detail below) to an alternative available channel in real-time to ensure that each aircraft maintains a reliable RF communications link. In one or more examples, if the pilot deviates from their advertised flight plan (for example by flying longer than anticipated) the spectrum management system **202** can be configured to take action (for instance by switching the RF channel) to ensure that any interruptions to the communications channel are mitigated.

[0144] In one or more examples of the disclosure, in addition to managing communication channels in real-time, the spectrum management system **202** can be configured to allocate and reserve one or more RF channels for a given flight to be used during the duration of the flight. As described in further detail below, the spectrum management system **202** can receive a flight plan from a pilot/operator **206** and based on the filed flight plan as well as other factors can allocate an RF channels to each flight in a deterministic manner that takes into account potential interference that may be encountered during the flight.

[0145] In one or more examples, and addition to assigning traffic channels to aircraft for communications with the ground, the spectrum management system **202** can also be configured to assign traffic channels to one or more ground base stations for the purpose broadcast/multi-case services that airborne radios in communication with the ground base station can utilize to gain access to services such as aircraft altimeter updates, enhanced GPS (RTK), and other aviation related services. In one or more examples, the spectrum management system **202** can also assign traffic channels to be allocated to the ground based station for dynamic allocation for on-demand request or traffic channels from airborne radios that are in communication with the ground based station but in idle mode on the ground station control channel. Examples of the use of these channels include aircraft that are operation on other networks (i.e., cellular, satellite, or other public/private terrestrial networks) that are looking to transition for such a service to another aviation network. In one or more examples, in order to facilitate the transition, the ground base station can dynamically allocate the sub-channel at the request of the pilot to the real-time spectrum management system **202** and the ground base station. Once assigned the traffic channel, the aircraft will continue flight operation on the allocated channel until the flight operation is complete or the channel is no longer needed (i.e., the aircraft has obtained service from another ground base station or other service provider). Once the channel is no longer needed, in one or more examples, the channel dynamically allocated to the aircraft can be returned to the dynamically allocated traffic channel pool for future dynamic assignment to another aircraft needing a dynamically allocated channel. In one or more examples, in the case of a dynamically allocated traffic channel as described above, the spectrum management system **202** can be aware of the flight plan filed for a given aircraft assigned a dynamic channel, even though the flight plan may be served by another network. In one or more example, the spectrum management system **202** can be aware of the location of the airborne radio of an aircraft through connectivity along a control channel that can continually monitor the location of the aircraft while continuously updating the spectrum management system **202**, as well as the beam forming antenna system both of which are prepared to provide a traffic channel upon request.

[0146] In one or more examples, the spectrum allocation process described above can be implemented by the spectrum management system **202**, or can be processed in a separate

component referred to herein as a “digital twin.” Due to the large volume of information and the potential for spectrum and/or traffic channel requests by tens of thousands of end users in the a given airspace, a digital twin of the spectrum management system can be used to perform the required analysis without impacting the operational system. In one or more examples, and as illustrated in the example of FIG. 2, the digital twin **204** can be implemented separately from the spectrum management system **202** so as to reduce the processing load of the spectrum management system **202**, thus leaving it free to perform the real-time operations associated with managing the active communications channels of aircraft transiting the airspace managed by the spectrum management system **202**. Alternatively, the digital twin **204** can also be implemented as part of the spectrum management system such that both the real-time management of air communication links, and flight planning are performed by the same component.

[0147] In one or more examples, the digital twin **204** can be configured to receive one or more requests from the pilots **206** for spectrum to use during a given flight plan. The digital twin, using the flight plan provided by the pilot as well as other factors (described below) can determine what RF spectrum channels to allocate to an aircraft when its flight commences. Once requests are confirmed in the digital twin **204**, execution and assignment of the communications channel on the operational spectrum management system **202** can be performed.

[0148] As described above, the spectrum management system **202** and the digital twin **204** can coordinate the RF spectrum needs of multiple aircraft in a given communications network so as to ensure that each individual aircraft can have access to a reliable and continuous communications channel with the ground during the entirety of its flight. In one or more examples, the spectrum management system **202** and the digital twin **204** can work in tandem to allocate and reserve RF spectrum channels for individual aircraft, and as described below, can monitor each individual communications link in-flight to ensure that the communications link is operating to its requirement.

[0149] Selecting an RF channel to allocate to a given flight can involve analyzing multiple variables to ensure that the selected channel will service the needs of an aircraft throughout the duration of its flight. In one or more examples, the spectrum management system **202** and the digital twin can analyze several variables such as the available spectrum resources, radio link throughput and performance requirements, location (including elevation), time-period as well as the radio frequency environment to assign a non-contended resource between the pilot and the aircraft. In one or more examples, the variables that influence channel selection can be populated by several internal and external components to the spectrum management system **202** that work together to match an aircraft to one or more RF channels for use during a flight as described below.

[0150] In one or more examples, the digital twin may use machine learning and/or predictive analytics to enhance its performance. In one or more examples, the digital twin may use machine learning and/or predictive analytics to predict the performance of aircrafts given the influence of external factors. The external factors may include weather, airspeed, barometric pressure, and/or other factors that may impact flight performance and thus the predictability of the network. In one or more examples, the digital twin may use machine learning and/or predictive analytics to predict network interference based on external factors or flight performance.

[0151] In one or more examples, each pilot in the communications network can interface with the communications network before and during their flight via the spectrum management system **202** and the digital twin **204**. Before the flight, and as described below, the pilot can interface with the spectrum management system and digital twin to receive an RF spectrum channel allocation for use during their flight based on their filed flight plan and other variables. During the flight, the spectrum management system **202** can provision the allocated RF spectrum channel to both the aircraft and the pilot to establish a continuous communications link and the spectrum management system can monitor the link during the flight to make sure it is performing within specification.

[0152] In one or more examples of the disclosure, the network **200** can include one or more base

stations that are not connected to the spectrum management system **202**. In one or more examples, a service provider who provides and maintains access to the spectrum management system **202** may not provide coverage to every desired geographic location. In one or more examples, in areas where a pilot may want to operate a flight but that does not fall within a coverage area of an existing base station, the service provider can provide the pilot with a temporary or portable base station **208**. In one or more examples, the temporary/portable base station may not have a connection with the spectrum management system **202** and thus cannot receive/transmit information to the spectrum management system for the purposes of provisioning RF channels to aircraft.

[0153] In one or more examples, the temporary/portable base station **208** can be used to setup point-to-point and multipoint links between the temporary/portable base station **208** and one or more aircrafts radios for flight operation. In one or more examples of the disclosure, the operator of a temporary/portable base station **208** the operators can inform the service provider a “concept of operation” of the base station **208** that describes the number of aircraft, the times they will fly and the spectrum they will use to communicate with the aircraft. While the spectrum monitoring system **202** may not send real-time information to the temporary/portable base station **208**, the spectrum management system **202** can use the concept of operation of the temporary/portable base station **208** to update the geofences (described in detail below) of the base stations **106a-c** that are connected to the network and can work to ensure that flights that are flying within its network **200** do not cause interference with the flight operations of the temporary/portable base station **208**. In one or more examples, the spectrum management system **202** can notify the operators of flights transiting the network **200** about the physical limitations to their operations caused by the temporary/portable base station **208** and can factor in the operations of the temporary/portable base station **208** when making RF spectrum slot allocations. In this way, while the spectrum management system **202** may not coordinate the operations of the temporary/portable base station **208**, it can work to protect its own network (i.e., the base stations that are connected to the spectrum management system) from the operations of the temporary/portable base station's point-to-point operations.

[0154] FIG. **3** illustrates an exemplary system for RF spectrum allocation and management according to examples of the disclosure. In one or more examples, the system **300** can represent a single link of a communications network shown in FIGS. **1-2**, and includes the components that manage the link between a pilot/end user **302** and an aircraft **336**. In one or more examples of the disclosure, the planning, creation, and operation of the link between the pilot **302** and the aircraft **336** can begin with the pilot **302** submitting information about their proposed flight to the digital twin **304**. In one or more examples, and as shown in FIG. **3**, the information transmitted by the pilot **302** to the digital twin **304** can include a flight plan, an aircraft/radio configuration, and a throughput requirement.

[0155] In one or more examples, the flight plan submitted by the pilot **302** can include flight mission details such as the intended timing, altitude, location, and speed of the aircraft during a proposed flight. In one or more examples, the pilot **302** can submit the flight plan to both a regulatory body (such as the Federal Aviation Administration (FAA)) for approval and additionally send the flight plan to the spectrum management system via the digital twin **304** for the purpose of obtaining one or RF spectrum channels for use during the proposed flight. In addition to the flight plan, the pilot **302** can send additional information to the digital twin **304** that the digital twin can use to select and allocate a RF spectrum channel to the user. For instance, in one or more examples, the pilot **304** can transmit the configuration of the aircraft or radio so as to inform the digital twin **304** as to the type of radio that the pilot will be communicating with during the flight. Knowledge of the radio configuration can allow for the digital twin **304** to not only understand the spectrum needs of the aircraft but can also allow for the digital twin to determine and predict other necessary information about the communications channel such as the modulation scheme and the forward

error correction code that will be active in-flight.

[0156] In one or more examples of the disclosure, the pilot **302** can also transmit a throughput requirement to the digital twin **304**. In one or more examples, the throughput requirement can represent the amount of data that is needed to be sent and received over the communications link. In one or more examples, the throughput can either be specified by the pilot **302** or can be derived based on the aircraft/radio configuration submitted by the pilot. For instance, in one or more examples, a particular aircraft (such as a UAV) may need a certain throughput of data for the channel to properly operate its autopilot features and thus by knowing the aircraft type, the system can derive the throughput requirements for that flight. As described in detail below, the throughput requirement can be used to determine the total amount of bandwidth for an RF spectrum channel and can thus inform the selection of a channel or channels that has an effective bandwidth to accommodate the throughput requirements of the flight.

[0157] As described above, the digital twin **104** can use the flight plan and other information transmitted to it by the pilot **302** as well as other information to select one or more RF spectrum channels for use by the pilot **302** during their flight. In one or more examples, the digital twin **104** can access a traffic channel pool **314** to determine the availability of RF spectrum channels to service a given flight. In one or more examples, the traffic channel pool **314** can represent all of the RF spectrum channels that could possibly be used to service a given flight. However, since there can be multiple aircraft in the network at any given time, and the need to reserve certain channels for emergency purposes (described in detail below), not every channel in the traffic channel pool **314** may be available for use by a particular aircraft during the times and locations required by a flight based on its flight plan.

[0158] FIG. **4A** illustrates an exemplary traffic channel pool according to examples of the disclosure. In one or more examples the traffic channel pool **400** can include a minimum frequency **402** and a maximum frequency **404**. The total range between the minimum frequency **402** and the maximum **404** can represent the total RF spectrum range of the network. In the example of FIG. **4A**, the minimum frequency is shown as 454.665 MHz while the maximum frequency **404** is shown as 454.985 MHz meaning that the total RF spectrum range is 320 kHz. Each and every flight operated by the communications network, must thus be allocated a channel within the 320 kHz frequency spectrum range. The frequency values shown in the example of FIG. **4A** are meant as examples only and should not be seen as limiting to the disclosure.

[0159] In one or more examples, the RF spectrum range created by the minimum frequency **402** and the maximum frequency **404** can be subdivided into one or more channels **406**. Each channel **406** can represent the smallest sub-portion of the RF spectrum range that can be allocated to an individual user. In the example of FIG. **4A**, the RF spectrum range of 320 kHz can be divided in 64 individual channels, with each channel represent a 5 kHz range. In one or more examples, the amount of channels that a flight needs for its intended flight plan can be based on the throughput requirements defined by the user as described above. For instance, in one or more examples, if a user requires a bandwidth that is greater than what 5 kHz can serve, then the system can allocate multiple channels **406** to service the flight. For instance, if a flight needed 15 kHz of bandwidth based on its throughput requirements, then the system can allocate three adjacent or non-adjacent 5 kHz channels in the traffic channel pool **400** so that the flight can have a bandwidth allocation commensurate with its throughput requirements. The channel sizes and frequencies described above and throughout the application are meant as examples only and should not be construed as limiting as the techniques described throughout can be applied to different frequencies and channel sizes and still fall within the scope of the disclosure.

[0160] As described above, given that there may be multiple flights operating on a network at any given time, not every channel in the RF spectrum range of the traffic channel pool **400** may be available to a given flight plan. Using the example of FIG. **4A**, one or more channels **408** may have already been previously allocated to other flight plans and thus may be unavailable for allocation to

a given user when deciding which channels to allocate. Additionally, in one or more examples, certain reserved channels **410** may be reserved for emergency uses and thus unavailable for allocation. For example, if during a flight a channel's performance degrades due to unforeseen circumstances, then the system may switch the flight from its allocated channel/channels to one of the reserved channels **408** so that it can maintain a reliable communication link with the ground. In one or more examples, the traffic channel pool **400** can include one or more restricted channels (not pictured) that cannot be used by the system because use of those signals is prohibited by various standards and rules that may be imposed by a regulatory authority. In one or more examples, the traffic channel pool **400** one or more channels that are designated as "in use" because the particular channel is either being used by a currently operating flight plan, or will be in use by another flight plan at the time that the current flight being planned will be in operation. In one or more examples, the traffic channel pool **400** can be designated as a "dynamically assigned" channel. As described above, a dynamically assigned channel can be allocated to the ground based station for dynamic allocation to aircraft to communicate with the ground based station even though the flight may not be actively using the network that the ground based station is operating on. Examples of the use of these channels include aircraft that are operation on other networks (i.e., cellular, satellite, or other public/private terrestrial networks) that are looking to transition for such a service to another aviation network. In one or more examples, in order to facilitate the transition, the ground base station can dynamically allocate the sub-channel at the request of the pilot to the real-time spectrum management system **202** and the ground base station. Once assigned the traffic channel, the aircraft will continue flight operation on the allocated channel until the flight operation is complete or the channel is no longer needed (i.e., the aircraft has obtained service from another ground base station or other service provider). Once the channel is no longer needed, in one or more examples, the channel dynamically allocated to the aircraft can be returned to the dynamically allocated traffic channel pool for future dynamic assignment to another aircraft needing a dynamically allocated channel. In one or more examples, in the case of a dynamically allocated traffic channel as described above, the spectrum management system **202** can be aware of the flight plan filed for a given aircraft assigned a dynamic channel, even though the flight plan may be served by another network. In one or more example, the spectrum management system **202** can be aware of the location of the airborne radio of an aircraft through connectivity along a control channel that can continually monitor the location of the aircraft while continuously updating the spectrum management system **202**, as well as the beam forming antenna system both of which are prepared to provide a traffic channel upon request. Thus, in one or more examples, the traffic channel pool **400** can have one more "dynamically assigned" channels that are reserved for use in the scenarios described above, and thus may not be available to be used for a flight that operating in the network. [0161] The example of FIG. **4A** described above provides an example of a traffic channel pool wherein each aircraft is assigned its own dedicated traffic channel for use to carry communications between a pilot and the UAS during a flight. However, this example should not be seen as limiting. For instance, in one or more examples, rather than assigning each UAS its own traffic channel, multiple UAS can share a single channel for instance by time multiplexing a channel amongst a plurality of UAS. FIG. **4B** illustrates another exemplary traffic channel pool according to examples of the disclosure. In one or more examples, the traffic channel **410** can include multiple sub-channels **416** similar to the example of FIG. **4A** and specifically channel **406** of FIG. **4A**. [0162] However, in contrast to the example of FIG. **4A**, rather than assigning each UAS assigned to a base station its own individual channel, in one or more examples, multiple UAS can be assigned to the same channel **416**. For instance, in one or more examples, and as illustrated in the example traffic channel pool **416** of FIG. **4B**, multiple UAS can share a single sub-channel **416** using a system in which the communications of each UAS assigned to the same sub-channel is time multiplexed. For instance, in one or more examples, a particular sub-channel **416** can have a pre-defined plurality of time slots.

[0163] Looking at the example of FIG. 4B, in one or more examples, a sub-channel **416** can be divided into multiple time slots **414**. In one or more examples, multiple UAS assigned to the same channel can transmit data during a time slot/slots allotted to them during a specific time. For instance, a sub-channel **416** (labeled SubCh 1, can be assigned to four different aircraft (UAS1-UAS4). In one or more examples, each aircraft can be assigned to a certain number of slots for a given sub-channel, for instance UAS 1 can be assigned slots **412A**, UAS2 can be assigned slots **412B**, UAS3 can be assigned slots **412C**, and UAS4 can be assigned slots **412D**. In one or more examples, each slot can be configured such that each UAS transmits once before any of the UAS assigned to a given sub-channel transmits for a second time in a given frame. In this way, each UAS assigned to a given sub-channel can transmit data within a given slot. For instance, UAS1-UAS4 can transmit data in a round-robin configuration such that each UAS1-UAS4 transmits during a slot before any of the UAS assigned to the channel transmits data during a second time slot.

[0164] In one or more examples, the size of a frame within a slot, and the size of the slots can be pre-defined. For instance, in one or more examples, the size of the frame can be pre-determined so as to optimize the balance between having a large frame that can improve channel performance (for instance to improve forward error correction (FEC) performance), and a small frame size that minimizes channel latency. In one or more examples, the frame size can also be determined by the hardware on either end of a communications channel. For instance, if a radio uses a frame size of 20 ms, then the frame size of the traffic channel pool **416** can be set to an integer multiple of the frame size of the radio (i.e., 80 ms). In one or more examples, the frame size and length of a time slot can be configurable by the user to meet the communication needs of the entity assigned to the slot. The example of FIG. 4B can thus allow for 4× the amount of UAS to use the traffic channel pool **410** in contrast with an example in which each sub-channel of the traffic channel pool is dedicated to a single aircraft. Similar to the example of FIG. 4A, one or more sub-channels **418** can be reserved (i.e., not assigned to any aircraft) to be kept in case of emergencies or for other reserved uses described above with respect to FIG. 4A. Furthermore, similar to the example of FIG. 4A, one or more sub-channels can also be assigned to a single UAS, while another sub-channel can be shared by multiple UAS.

[0165] In one or more examples, the traffic channel pool **416** of FIG. 4B can represent a forward slot structure associated with transmissions from the ground to the aircraft. In one or more examples, the return slot structure (i.e., transmissions from the aircraft to the ground) can be almost identical to the forward frame structure except the return traffic associated with a particular aircraft can be allocated a different time slot than the time slot allocated to the aircraft in the forward frame in order to allow for half-duplex operation of the aircraft's radio. In one or more examples, the return slots can also be placed into a different RF sub-channel than the forward slots. In one or more examples, allowing a single RF channel to be divided in time by multiple aircraft can lead to increased spectral efficiency by allowing more aircraft to use the RF sub-channels of the traffic channel pool than if a single aircraft were assigned to its own dedicated RF sub-channel.

[0166] Returning to the example of FIG. 3, the digital twin **304** can select a channel or channels from the traffic channel pool **314**, which as described above can include available sub-channels **316**, reserved channels **318**, and restricted traffic channels **320**. However, in one or more examples, simply selecting an available sub-channel or channels for allocation may not mean that a communication link using that channel will be reliable throughout a flight. Various factors such as the predicted RF environment experienced during a flight or predicted interference can cause a given channel to fail during a flight. Thus, as part of the allocation process, the digital twin **304** can also perform an analysis to ensure that the channel will be available and perform to requirements throughout the duration of the flight as described below.

[0167] In order to allocate an RF channel to an aircraft, the digital twin **304**, in one or more examples, can first determine if RF coverage is available for the aircraft during the entirety of its

flight. To do so, in one or more examples, the digital twin **104** of the spectrum management system can “geofence” the coverage areas of each of the ground based stations in the network as shown at **306**. In one or more examples, a “geofence” **306** can refer to a zone within a coverage area in which there is sufficient RF availability for flight traffic. In one or more examples, when a pilot **302** submits a flight plan, the system can query the geofences **306** to make sure there is RF availability throughout the path of the plan and at all altitudes expressed in the flight plan. In one or more examples of the disclosure, geofences can be shared with the pilot/operator of a flight and can be programmed into the auto-pilot of an aerial vehicle for use during a flight.

[0168] In one or more examples, a geofence can be created using a dynamic link budget **308** that is maintained by the digital twin **304**. In one or more examples, each geofence **306** can have its own dynamic link budget **308**. The dynamic link budget **308** can determine what the RF availability of a given geofence is at any particular moment in time, and can even predict RF availability for a given geofence in the future based on various parameters. In one or more examples, the dynamic link budget **308** can include such parameters as antenna gain, RF losses, receiver sensitivity, power, frequency, spectrum bandwidth, traffic channel size/quantities (i.e., sub-channels, resource blocks), quality of service (QOS) requirements, modulation, spectrum monitoring system results (described in further detail below) and the location of any known co-channel interferers. The dynamic link budget **308** can also include an RF safety margin to ensure a reliable communications signal in the geofence **306**. In one or more examples, the operational spectrum management system **322** (described in detail below) can maintain a real-time version of the link budget that changes based on changing conditions in the RF environment. In one or more examples, the digital twin **304** can maintain a model of the link budget, and that dynamic link budget **308** can be used to predict RF conditions at a future time based on the times implicated by a given flight path. In one or more examples, the dynamic link budgets for each geofence can be validated using measurements of RF spectrum activity at each of the base stations in the area to ensure that the dynamic link budgets include the most up-to-date information and accurately reflect the RF environments that the dynamic link budgets are meant to model. In one or more examples, each geofence can be configured to predict coverage based upon components of the flight plans presented to the spectrum management system, the spectrum monitoring systems employed at each base states, the capabilities of the beam/null forming antennas at each base station and/or satellite, as well as the known locations of other airborne radios. In one or more examples, the actual performance of radio links created at the base stations can be monitored and the information sent to the spectrum managements system for validation and modification of the geofences.

[0169] In one or more examples, and as part of the process of allocating RF spectrum channels to an aircraft, the digital twin **104** can cross reference the dynamic link budget with a calibrated RF coverage prediction tool **310**. In one or more examples, the RF coverage prediction tool **310** uses appropriate RF prediction models, morphology, topology, antenna pattern characteristics, and antenna elevations to create the dynamic geofence coverage area based upon the remote radio configuration and user requirements. In one or more examples, the RF coverage prediction tool **310** may use machine learning and/or predictive analytics to create the dynamic geofence coverage area. In one or more examples, the RF coverage prediction tool **310** can be used to generate a dynamic link budget for each geofence coverage area that a flight will transit based on its filed flight plan. In one or more examples, if it is determined via the dynamic link budget/budgets that a given flight plan may not be able to sustain a reliable communications channel throughout its flight, then the pilot **302** can be notified that their flight plan must be altered in order to give the aircraft **336** and pilot **302** the communications channel needed for the duration the flight. In one or more examples, if it is determined that the flight plan is serviceable, then in one or more examples of the disclosure, the specific RF spectrum channel or channels allocated to the flight from the traffic channel pool can be cross-referenced against a dynamic interference and co-existence prediction tool **312** (“interference tool”) to determine if the particular frequency allocated to the

flight as described above has the potential to interfere or be interfered with by another flight in the network. In one or more examples, the interference tool **312** can be configured to calculate known co-channel interferences that may occur during a flight. In one or more examples, co-channel interference can be caused by other remote radios operating in the network that can be distributed geographically and in altitude across the geographic coverage areas that a given flight will transit based on its flight plan. If it is determined that a co-channel interference could render a given channel allocation, then in one or more examples, the digital twin **304** can select another channel or channels from the traffic channel pool and analyze the allocated channel using the interference tool **312** to determine if the channel will be reliable and available throughout the flight plan as transmitted by the pilot **302**. In one or more examples, the spectrum management system can use the interference tool **312** to set one or more exclusion zones (i.e., where aircraft are not allowed to fly). Additionally, the spectrum management system can set one or more exclusion zones based on various regulatory requirements, interference, point-to-point operations, and alternative technology operations including satellite or terrestrial communications networks.

[0170] In one or more examples, the spectrum management system may determine the handover points between each geofence coverage area that is involved in the course of a flight. In one or more examples, a handover point can refer to a time or location at which transmission to/from an aircraft is transferred from one base station or satellite to another base station or satellite without the aircraft losing connectivity. The location and elevation of the handover points may be calculated by the RF coverage prediction tool **310** before a flight. In one or more examples, the information related to the predicted handover points may be made known to the pilot before the flight. The pilot may use the information related to the predicted handover points to assist in the process of determining exclusion zones, clearing conflicts in resource allocations, and configuring the equipment and antenna systems in the related geofence coverage areas. In one or more examples, the information related to the predicted handover points may be programmed into the autopilots such that the aircraft and the pilots are made aware of the handover activity before the completion of the flight.

[0171] Thus, as described above, the dynamic link budget **308** (in conjunction with the dynamic RF coverage prediction tool **310**) can be configured to determine if a given flight plan will have RF coverage at all points and times during the flight plan, while the interference tool **312** can be configured to ensure that the channel allocated from the traffic channel pool **314** will not be subject to a harmful amount of interference during the flight.

[0172] In one or more examples, the spectrum management system may interact with other spectrum management systems to form a network of networks to service a specific flight plan. In one or more examples, the coordination of multiple spectrum management systems may be performed by an airspace integrator. The airspace integrator may allow multiple spectrum management systems to work efficiently together to solve the coverage requirements of a flight plan when coverage with one specific wireless service with a specific spectrum management system is unable to meet the needs of the requested flight plan. In one or more examples, the airspace integrator may include information from global stakeholders (e.g., air traffic regulators, military flight operators, aerospace administrators, etc.), weather services, civil rules related to countries' aviation spaces, and unmanned traffic management systems. The airspace integrator may use the integrated information to allocate physical space for aircrafts, as well as assist the wireless network operators with their spectrum management systems to ensure predictable flight performances based on submitted flight plans. A detailed description of an exemplary airspace integrator is provided further below.

[0173] As described above, if the network of networks, dynamic link budget **308**, or the interference tool **312** determine that a reliable RF link cannot be established during the flight, or that a channel that meets the needs of the flight plan is not available during the proposed time of the flight, then in one or more examples, the digital twin **304** can inform the pilot **302** that the flight

plan needs to be adjusted.

[0174] FIG. 5 illustrates an exemplary channel allocation process for RF spectrum management of ground to air communication networks according to examples of the disclosure. The process 500 of FIG. 5 can illustrate the interactions between a pilot 302 and the digital twin 304 to receive an RF spectrum channel allocation as described above. In one or more examples, the process 500 can begin at step 502 wherein the system (via the digital twin 304) can receive flight information from a pilot or user of the system who wants to reserve an RF spectrum channel or channels for a given flight. As described above, and referring to FIG. 3, the flight information received at step 504 can include a flight plan, an aircraft/radio configuration, and a throughput requirement. Thus at step 502, the digital twin 304 can receive the information from the pilot/operator that is necessary to determine the RF channel needs of a given flight as well as the geographic locations and altitudes that the flight will occupy during its flight.

[0175] In one or more examples, once the information is received at step 502, the process 500 can move to step 504 wherein the received flight information can be mapped against site coverage and available channel capacity. In one or more examples, at step 504, the digital twin 304 can use the information received from the pilot 302 to determine if there is RF coverage for the entirety of the proposed flight (as determined by the dynamic link budget 308), the digital twin 304 can also allocate an available channel or channels from the traffic pool 314, and can determine if there will be any potential interference with the channel using the interference tool 312. In one or more examples, determining RF coverage for the entirety of the flight can include making sure that for each RF channel to be utilized by a flight plan, there are available time slots on the channel to facilitate transmissions from the aircraft associated with the flight plan. In one or more examples, determining RF coverage can include taking into account any factor that can have an effect on RF service such as sources of interferences. Thus, in one or more examples, mapping a flight plan against site coverage can not only include determining whether RF traffic resources are available for the flight, but can also include determining whether the quality of RF service during the flight meets an acceptable minimum threshold.

[0176] In one or more examples, once the flight has been mapped against site coverage and available channel capacity in order to select a channel or channels for use during the flight at step 504, the process 500 can move to step 506 wherein the availability of the channel is confirmed and an approval for the flight plan with channel allocation is requested. In one or more examples, a regulatory body such as the Federal Aviation Administration (FAA) may be required to approve any flight plans including channel allocations. Thus at step 506, once a channel has been allocated and the availability of the channel has been confirmed, the flight plan with spectrum allocation can be transmitted to the regulatory body for approval if required.

[0177] After mapping the flight against site coverage and available channel capacity in step 504, if in step 506 the system cannot confirm channel availability, in one or more examples, the system can reject the received flight plan and alert the user that the flight plan has been rejected.

Additionally or alternatively, in one or more examples, in addition to rejecting a flight plan based on lack of channel availability, the system can provide the user with a suggested modified plan that allows the plane to travel from its intended origin to its intended destination (based on the user submitted flight plan) in a manner that will ensure RF channel availability throughout the flight. In one or more examples, modifying the flight plan can include modifying the intended geographic route of the aircraft (e.g., location and altitude information) based on the class of airspace the flight is being conducted in, based on the RF coverage availability of the communications network, or based on traffic channel availability of the communications network. Additionally or alternatively, in place of rerouting the flight, in one or more examples, the system can modify the time window (i.e., the time the flight can take off) in which the flight can be conducted based on traffic channel availability. Alternatively, in one or more examples, depending on the priority of the flight, instead of modifying the flight plan, the system can modify the ability of the communications network to

support the flight plan (e.g., by changing or canceling another flight plan in the communications network based on its priority and alerting the user associated with that flight that that their flight plan has been changed or canceled). For instance, if the flight plan is submitted to the system by a user with high priority (for instance for a medical flight) then in one or more examples, the system can modify or cancel the flight plans of other lower priority flights in order to ensure RF channel availability for the priority flight during the flight proposed in the submitted flight plan for the priority flight.

[0178] In one or more examples, once the request for approval has been transmitted at step **506**, the process **500** can move to step **508** wherein the approval from the regulatory body can be received. Additionally, at step **508**, once the approval has been received, then the system (i.e., the digital twin **304**, can reserve the channel or channels that were originally allocated to the flight and approved by the regulatory body. In one or more examples, once the channel has been reserved at step **508**, the process **500** can move to step **510** wherein the reserved channel is assigned to the pilot and the aircraft for use when the flight plan is to be executed.

[0179] Referring back to FIG. 3, and as described above, the digital twin **304** can be responsible for allocating spectrum and planning operations for a flight, before the flight is to take place. However, the actual operations of the communications channel can be handled by a separate spectrum management system **322**. In one or more examples, the digital twin **304** and the spectrum management system **304** can be implemented as a single system. Alternatively, the digital twin **304** and the spectrum management system **322** can be implemented as separate systems. In one or more examples of the disclosure, the spectrum management system **322** can be responsible for managing all of the communications link that are operating in a given communications network. Because the spectrum management system **322** is operating in real-time and must make decisions that can affect multiple communications link, in one or more examples, it can be advantageous to implement the digital twin **304** and the spectrum management system **322** on separate systems such that the operations of the digital twin **304** will not impact the speed at which the spectrum management system **322** performs its operations. In one or more examples, and as described below, the spectrum management **322** can be responsible for implementing and managing the communications links for all flights in a given air-to-ground communications network. Thus, once the spectrum requests are confirmed by the digital twin **304**, execution and assignment of the communications channel with the allocated channel or channels can be performed on the spectrum management system **322**.

Referring back to the example of FIG. 5, once the channel or channels have been assigned to the operator by the digital twin **304** at step **510**, the process **500** can move to step **510** wherein the channel allocation and other additional information about the channel can be sent to the spectrum management system **322** for implementation during flight. In one or more examples, the additional information can include other parameters regarding the communications channel to be predicted by the spectrum management system **322** such as the modulation scheme, error correction codes etc. Additionally, or alternatively, the rather than the digital twin **304** transmit the additional information, the spectrum management system **322** itself can determine the additional information needed to establish a given communications during the flight.

[0180] As described above, the spectrum management system **322** can be responsible for not only implementing all of the communication channels between the pilots **302** and the flight **336** in a given communications network, but can also be responsible for monitoring the links in real-time to ensure that they are operating according to their requirements during flight. In order to perform its implementation and monitoring tasks, in one or more examples, the spectrum management system **322** can be in communication with the each of the base stations **326** of the communications network via the base station controller **324** described above with respect to FIGS. 1-2. In one or more examples of the disclosure, each base station in the network can include one or more components and tools to assist the spectrum management system **322** in setting up and monitoring communication links in real-time as aircraft **336** transit the coverage area of the base station. The

base station controller **324** can be communicatively coupled to the base station controller as to access the components and tools necessary to implement and monitor the communication links that the spectrum management system is charged with setting up and maintaining such as one or more antenna elements and tools used to monitor the RF environment of a given base station.

[0181] In one or more examples of the disclosure, the spectrum management system **322** can have access to and control a spectrum monitoring device **328** located at each base station **326** in the communications network. In one or more examples, the spectrum monitoring device can include one or more hardware components (such as antennas and sensors) that are collectively configured to monitor the RF environment of the base station **326**. A spectrum monitoring device can be located at each and every base station in the communications network and can be configured to continuously measure the active RF environment of a base station for interference.

[0182] FIG. **6** illustrates an exemplary spectrum monitoring device according examples of the disclosure. In one or more examples the spectrum monitoring device **604** can be implemented in the base station controller **324** which can communicate information about the spectrum to the digital twin and the spectrum management updates so as to automatically and in real-time update the geofences managed by the spectrum management system. The base station controller **324** can be communicatively coupled (for example by a coax connection) to one or more tower mounted monitoring antennas **602** that can act as the sensors that the spectrum monitoring device uses to monitor the RF environment of the base station. In one or more examples, the antennas **602** may be mounted on a satellite. In one or more examples, the spectrum monitoring device **604** can include a software defined receiver **606** that can be configured to receive and process RF signals received by the monitoring antennas **602**. The software defined receiver can be used by one or more components of the spectrum monitoring device **604** to perform the analysis required to monitor the RF environment of the base station.

[0183] In addition to detecting unwanted RF signals, the spectrum monitoring device **604** can include a direction/power detection component **612** that can detect the precise power and direction of the unwanted signal. As described in detail below, the base station can include a beam steering antenna that can be utilized to minimize or eliminate unwanted and potentially interfering RF signals. Thus, the direction/power detection component **612** can be used to determine at what power and direction the unwanted signal is coming from and can use a beam steering antenna to nullify or minimize the unwanted signal. In one or more examples, the beam steering antenna may be configured with an individual radio frequency front end for each individual traffic/resource channel. The radio frequency front ends may dynamically or directly adjust the power from the antenna towards the desired or undesired target to improve radio performance, reduce noise, increase radio link performance or system capacity, and provide additional power management capabilities. In one or more examples, the direction/power detection component **612** and beam steering antenna may be mounted on a satellite.

[0184] In one or more examples, the spectrum monitoring device **604** can include a RF signature database **614**. The RF signature database **614** can allow for the spectrum monitoring device to compare any identified and unwanted RF signals against a database of known RF signature for identification. In one or more examples, if the RF signature of an unwanted interferer (such as a malicious user who is using the spectrum without permission) can be identified using the RF signature database, then the incident can be reported to a regulatory authority for potential action against the malicious user.

[0185] In one or more examples, the spectrum management system **322** can utilize the data generated by the spectrum monitoring device **604** to adjust the one or more communication channels in its charge so as to ensure that each communication link performs to its desired level of performance. Referring back to FIG. **3**, the base station **326** can include one or more beam steering antenna components **330** configured to mitigate or eliminate unwanted RF signals from the base station coverage area. In one or more examples, RF interference can come from a known

interference source such as a high altitude aircraft flying over an adjacent airspace, or from an uncooperative source such as a malicious user who is using the RF spectrum without permission. As a defense against these types of interferences, in one or more examples, the base station **326** can include a beam/null forming steering antenna **330** that can point a null at the interferer and work to either eliminate or substantially reduce the interference that the interferer may be causing to aircraft operating at the base station.

[0186] FIG. 7 illustrates an exemplary beam steering antenna system according to examples of the disclosure. FIG. 7 illustrates the exemplary beam steering antenna system in the context of a communications network **700** to better illustrate the features of the system. In one or more examples, communication network **700** can include two separate base stations **708** and base station **704**. In one or more examples, base station **708** can transmit a desired signal **714** to an aircraft **706** that is transiting the coverage area of base station **708**. In the example of FIG. 7, base station **708** is illustrated as transmitting an RF signal of 459.825 MHz. In one or more examples, the aircraft **706** can be flying at an altitude of 25,000 feet as shown in the figure.

[0187] In one or more examples, communications system **700** can also include a base station **704** that is operating in a coverage area that is adjacent to the coverage area of base station **708**. In one or more examples, base station **704** can transmit a desired signal **712** to an aircraft **702** transiting its own coverage area at the same time that aircraft **706** is transmitting the coverage area of base station **708**. In the example of FIG. 7, aircraft **702** is shown to be flying at 1,200 feet and communicating with base station **704** using an RF channel centered at 459.825 MHz, which is the same frequency that aircraft **706** is using to communicate with its corresponding base station **708**.

[0188] As shown in FIG. 7, though aircraft **706** is operating at a different base station than aircraft **702**, its communications with the base station **708** can cause interference with the communications of aircraft **702** with its base station **702**. In one or more examples, owing to its relatively high altitude, the desired signal **714** of base station **708** (which is directed toward aircraft **706**) can be seen by base station **704** in the form of undesired signal **710**. Undesired signal **710**, as it is caused by the communications between base station **708** and aircraft **706**, can be at 459.825 MHz, which is the same frequency as signal **712** between aircraft **702** and base station **704**. Thus, the undesired signal **710** can cause interference with the desired signal **712**.

[0189] In one or more examples, using the spectrum monitoring device described above with respect to FIG. 6 as well as a base station link monitoring tool (described in detail below), the spectrum management system **322** knowledge of all known airborne radios on the network can validate or detect the presence of a known or unknown undesired signal **710** and utilize a beam/null forming steering antenna **716** that is mounted to a tower of the base station **704** to substantially reduce or eliminate the undesired signal **710** from the RF spectrum environment of aircraft **702**. In one or more examples, beam/null forming steering antenna **716** may be mounted on a satellite. In one or more examples, the beam/null forming steering antenna **716** can be configured as a steerable antenna that can be pointed in the direction of an undesired signal. In one or more examples, the beam/null forming steering antenna **716** can be implemented as a phased array antenna with beam forming functionality that can be configured to transmit RF energy in a specific desired direction. In one or more examples, the direction of the beam can be controlled by a base station controller connected to the base station **704** or by the spectrum management system **322** directly. In one or more examples, if the beam/null forming steering antenna **716** is mounted on a satellite, the base station controller can communicate with the satellite to control the antenna to cause it to control the direction of the beam.

[0190] In one or more examples, if the spectrum management system **322** detects an undesired signal at a particular base station in the network, the spectrum management system **322**, using the spectrum monitoring device, can determine the direction of the signal as well as the power of the signal (for instance by using direction/power detection component **612**). Once the spectrum management system **322** has determined the power and direction of the undesired signal it can

control the beam steering antenna **716** to form a null in the direction of the undesired signal and with the introduction of noise cancelling techniques like Successive Noise Cancellation and Multi User Detection together can substantially reduce the impact of the undesired signal in the RF coverage area of the base station **704**. In one or more examples, the beam may be configured to transmit RF energy in a specified direction to cancel noise between desired and undesired targets. [0191] Referring back to FIG. **3**, in addition to the spectrum monitoring device **328**, and the beam/null forming steering antenna **330**, in one or more examples, the base station **326** includes a base station link monitoring functionality **332**. In one or more examples, the base station link monitoring functionality **332** monitors each of the individual links that are assigned by the spectrum management system **322** at the base station **326**. While the spectrum monitoring functionality **328** is configured to monitor all the active radio links individually of the base station **326**, and can update real time the spectrum management system **322** to validate and adjust as needed the individual link predicted performance and utilization by the aircraft and operator. [0192] In one or more examples, if the spectrum management system **322** detects conditions or interference of all communication links in the network, then it can work to mitigate any degradation in performance by adjusting the dynamic link budget, updating the dynamic RF coverage predictions implementing the beam/null forming steering antenna capabilities to apply a null at a detected source of interference and notify the operator of any coverage change. However, in one or more examples, if the spectrum management system **322** is unable to mitigate a situation through the techniques describe above, then the spectrum management system **322** can change the RF spectrum channel allocation of an individual aircraft in the network so as to find a more favorable communications channel for use. Thus, in one or more examples, the base station can include one or more sub-channel and resource blocks **334** that are reserved by the system to allocate to aircraft in-flight who may be experiencing service issues and need to change their frequency.

[0193] As described above with respect to FIG. **4A**, one or more channels in the traffic channel pool **314** can be reserved for emergency purposes. The base station **326** being controlled by spectrum management system **322** can have access to these reserved channels for the purpose of allocating those channels to an aircraft mid-flight in case the originally allocated communications link fails or has an unacceptable quality of service. In one or more examples, the base station controller **324** can also be configured to operate the link once the flight is airborne. Additionally, the base station controller **324** can work during the flight to ensure that a quality RF communications link is provided to each and every aircraft in the communication network. Thus, when a pilot is ready to take off and execute their flight plan, in one or more examples the base station controller **324** can implement and manage the communication link for the duration of the flight using the parameters provided to it by spectrum management system **322**. In one or more examples of the disclosure, the spectrum management system **322** can track the flight as it is occurring (using data from the base station controller **324**) and can compare the flight plan submitted by the user to the actual flight. Once the flight has concluded, in one or more examples, the spectrum management system can notify the base station controller when the link is no longer needed, by deregistering the airborne radio of an aircraft from the network and returning the allocated RF spectrum channel to the traffic channel pool **314**.

[0194] FIG. **8** illustrates an exemplary in-flight process for RF spectrum management of ground to air communication networks according to example of the disclosure. In one or more examples, the process **800** of FIG. **8** can begin at step **802** wherein the channel information pertaining to a flight is received from the spectrum management system **322** at a base station controller **324**. As discussed above, the digital twin **304** of FIG. **3** can be configured to ingest a flight plan and other data from a pilot and allocate an RF spectrum channel or channels for a traffic channel pool **314** based on the provided information. Once that channel has been allocated and other parameters regarding the channel have been determined, the digital twin **304** can transmit the information,

which is then received by the spectrum management system **322** and transmitted to the base station controller **324** at step **802**. The information provided from the spectrum management system **322** to the base station controller **324** can inform the base station control **324** as to how to provision a communication channels for a particular flight at the time the flight **336** is executed by the pilot **302**.

[0195] When the flight is ready to commence per its flight plan, in one or more examples, the process **800** can move to step **804** wherein the base station controller **324** creates the communication link (using a beam forming antenna) using the communication link parameters transmitted by the spectrum management system **322** (which were generated by the digital twin **304**). In one or more examples, creating the communication link can include pointing antennas in the appropriate direction (where the flight will be), setting the proper RF frequencies for transmitting and receiving signals at the base stations based on the allocated RF spectrum channel, and applying any other parameters such as modulation scheme and forward error correction code.

[0196] In one or more examples, once the link has been created at step **804** and the flight is in progress and is utilizing the communications link, the process **800** can move to step **806** wherein the communications is monitored by the base station controller **324** that is connected to the base station **326**. In one or more examples, monitoring the communications link can include monitoring of the spectral environment of the base station using a spectrum monitor device **328** and/or a base station link monitoring tool **332**, as described above. At step **804**, the process **800** can monitor the link to ensure that the requirements of the link are being met at all times. As part of monitoring the link at step **804**, the spectrum management system **322** can create, maintain, and update one or more geofences of the coverage areas associated with a base station by updating a real-time dynamic link budget, as well as coverage and interference applications for the flight. In one or more examples, the dynamic link budget can include the same parameters as the dynamic link budget kept by the digital twin, however rather than serving to predict future link performance, the dynamic link budget maintained by the spectrum management system can be updated in real-time to monitor link performance during the flight. In one or more examples, the dynamic link budget can be updated using information provided to the spectrum monitoring device **328** and the base station link monitoring tool **332**.

[0197] Once the monitoring begins at step **806**, the process **800** can move to step **808** wherein a determination is made as to whether the communications link is being interfered with. In one or more examples, the interference can be detected by changes to the dynamic link budget or directly from the spectrum monitoring device **329** and/or link monitoring tool **332**. In one or more examples, if no interference is detected at step **808**, the process **800** can revert back to step **806** wherein the communications link is further monitored for interference. If, however, at step **808** interference is detected, then the process **800** can move to step **810** wherein the spectrum management system or the base station controller attempts to mitigate the interference. In one or more examples, mitigating the interference can include adjusting the Geofence that includes the dynamic link budget, changing traffic channels, and/or notifying pilot of available geofenced service area. Additionally or alternatively, interference can be mitigated using the beam steering antenna described above with respect to FIG. 7 to apply a null signal in the direction and frequency of the interferer to try and mitigate its effect on the communications link of the flight.

[0198] In one or more examples, and after the mitigation has been applied at step **810**, the process **800** can move to step **812** wherein a determination is made as to whether the issue that prompted the mitigation at step **810** has resolved the issue. In one or more examples of the disclosure, determining whether the issue has resolved can include looking to the dynamic link budget (as updated in response to the mitigation) to see if the communications link is now operating according to its requirements. In one or more examples, if the issue has been resolved as determined at step **812**, the process **800** can revert back to step **806** wherein the link is further monitored.

[0199] If, however, the issue is not resolved as determined at step **812**, the process **800** can move to

step **814** wherein the spectrum management system **322** can change the channel allocated to the flight for instance by allocating one of the reserved channels to the flight from the traffic channel pool. As described above, the channels that are reserved from the traffic channel pool for emergency uses can be used in situations where the allocated channel for a flight is no longer serviceable, or in situations where the flight plan is changed (for example by changing how long the flight will be in the air) such that the channel it was originally allocated may not be available for the entire flight. Thus, in one or more examples, at step **814** the process **800** can change the allocated channel of the flight to a reserved channel in an attempt to still provide the flight with a reliable and continuous channel. In one or more examples, as part of the process of changing the channel at step **814**, the spectrum management system **322** can inform the pilot as well as the aircraft about the change.

[0200] As described above, the spectrum management system **322** can setup and monitor a communications link for the duration of a flight, and has the capability to take action should a reliable and continuous communication link become threatened during a flight. In conjunction with the digital twin **304**, the entire system can plan a communications channel for flight, implement the communications for the flight, and respond to issues with the communications links of flight during the flight.

[0201] As described above, spectrum management system **322** may interact with other spectrum management systems to form a network of networks that works efficiently to solve the coverage requirements to service a specific flight plan. A network of networks could coordinate coverage from one or more wireless network operators when a single spectrum management system is unable to meet the needs of the requested flight plan. A network of networks may also coordinate multiple spectrum management systems to allow for the best wireless connectivity to be utilized during a flight to provide more reliable network connection over a greater geographical area.

[0202] FIG. **9** illustrates an exemplary system for network of networks according to examples of the disclosure. The network of networks **900** may include one or more wireless network operators **902**. In one or more examples, each network in the one or more wireless network operators **902** may include a spectrum management system with a digital twin that operates according the descriptions above.

[0203] The network of networks **900** may include pilot **904**. Pilot **904** may be an example of pilot **304** shown in FIG. **3**. Pilot **904** may provide inputs to the network of networks **900**. In one or more examples, the pilot **904** may submit flight plans, configure aircrafts and radios, and/or input throughput requirements. Input from pilot **904** may be used by airspace integrator **906** to implement communication channels of wireless network operators **902**.

[0204] In one or more examples, the coordination of multiple wireless network operators **902** may be performed by an airspace integrator **906**. Airspace integrator **906** may allow multiple wireless network operators **902** to work efficiently together to solve the coverage requirements of a flight plan when coverage with one specific wireless service with a specific spectrum management system is unable to meet the needs of the requested flight plan.

[0205] In one or more examples, airspace integrator **906** may include information from global stakeholders **910**. Global stakeholders **910** may include air traffic controls, NASA, national regulators, military flight operators, aerospace integrators, weather services, EASA, and unmanned traffic management systems. In one or more examples, airspace integrator **906** may include rules related to countries' civil airspace **908**. The airspace integrator **906** may use the integrated information from global industry stakeholder **910** and civil airspace **908** to allocate physical space for aircrafts, as well as assist the wireless network operators **902** with their spectrum management systems to ensure predictable flight performances based on submitted flight plans.

[0206] FIG. **10** illustrates an example of a computing system **1000**, in accordance with some embodiments. System **1000** can be a client or a server. As shown in FIG. **10**, system **1000** can be any suitable type of processor-based system, such as a personal computer, workstation, server,

handheld computing device (portable electronic device) such as a phone or tablet, or dedicated device. The system **1000** can include, for example, one or more of input device **1020**, output device **1030**, one or more processors **1010**, storage **1040**, and communication device **1060**. Input device **1020** and output device **1030** can generally correspond to those described above and can either be connectable or integrated with the computer.

[0207] Input device **1020** can be any suitable device that provides input, such as a touch screen, keyboard or keypad, mouse, gesture recognition component of a virtual/augmented reality system, or voice-recognition device. Output device **1030** can be or include any suitable device that provides output, such as a display, touch screen, haptics device, virtual/augmented reality display, or speaker.

[0208] Storage **1040** can be any suitable device that provides storage, such as an electrical, magnetic, or optical memory including a RAM, cache, hard drive, removable storage disk, or other non-transitory computer readable medium. Communication device **1060** can include any suitable device capable of transmitting and receiving signals over a network, such as a network interface chip or device. The components of the computing system **1000** can be connected in any suitable manner, such as via a physical bus or wirelessly.

[0209] Processor(s) **1010** can be any suitable processor or combination of processors, including any of, or any combination of, a central processing unit (CPU), field programmable gate array (FPGA), and application-specific integrated circuit (ASIC). Software **1050**, which can be stored in storage **1040** and executed by one or more processors **1010**, can include, for example, the programming that embodies the functionality or portions of the functionality of the present disclosure (e.g., as embodied in the devices as described above)

[0210] Software **1050** can also be stored and/or transported within any non-transitory computer-readable storage medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a computer-readable storage medium can be any medium, such as storage **1040**, that can contain or store programming for use by or in connection with an instruction execution system, apparatus, or device.

[0211] Software **1050** can also be propagated within any transport medium for use by or in connection with an instruction execution system, apparatus, or device, such as those described above, that can fetch instructions associated with the software from the instruction execution system, apparatus, or device and execute the instructions. In the context of this disclosure, a transport medium can be any medium that can communicate, propagate or transport programming for use by or in connection with an instruction execution system, apparatus, or device. The transport computer readable medium can include, but is not limited to, an electronic, magnetic, optical, electromagnetic, or infrared wired or wireless propagation medium.

[0212] System **1000** may be connected to a network, which can be any suitable type of interconnected communication system. The network can implement any suitable communications protocol and can be secured by any suitable security protocol. The network can comprise network links of any suitable arrangement that can implement the transmission and reception of network signals, such as wireless network connections, T1 or T3 lines, cable networks, DSL, or telephone lines.

[0213] System **1000** can implement any operating system suitable for operating on the network. Software **1050** can be written in any suitable programming language, such as C, C++, Java, or Python. In various embodiments, application software embodying the functionality of the present disclosure can be deployed in different configurations, such as in a client/server arrangement or through a Web browser as a Web-based application or Web service, for example.

[0214] The foregoing description, for the purpose of explanation, has been described with reference to specific embodiments. However, the illustrative discussions above are not intended to be

exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in view of the above teachings. The embodiments were chosen and described in order to best explain the principles of the techniques and their practical applications. Others skilled in the art are thereby enabled to best utilize the techniques and various embodiments with various modifications as are suited to the particular use contemplated. For the purpose of clarity and a concise description, features are described herein as part of the same or separate embodiments; however, it will be appreciated that the scope of the disclosure includes embodiments having combinations of all or some of the features described.

[0215] Although the disclosure and examples have been fully described with reference to the accompanying figures, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosure and examples as defined by the claims. Finally, the entire disclosure of the patents and publications referred to in this application are hereby incorporated herein by reference.

Claims

1. (canceled)

2. A method for allocating communication resources in an air-to-ground communication network of networks, the method comprising: receiving operational flight requirements associated with a plurality of flight operations that are to fly in a coverage area of an air-to-ground communications network; determining communication resource availability for a flight that is to fly in the coverage area of the air-to-ground communications network, wherein determining communication resource availability for the flight comprises: receiving a traffic resource pool from a communication network, wherein the traffic resource pool includes one or more communication resources; and selecting a communication resource from the received traffic resource pool, wherein selecting the communication resource from the received traffic resource pool is based on the received operational flight requirements associated with the plurality of flight operations; determining that a performance of the selected communication resource is above a pre-determined threshold, wherein the determination is based on a radio configuration information associated with the flight that is to fly in the coverage area; and in accordance with the determination that the performance of the selected communication resources is above the pre-determined threshold, operating the flight using the selected communication resource from the received traffic resource pool.

3. The method of claim 2, wherein determining communication resource availability in the coverage area of the air-to-ground communications network based on the received flight plan from the user further comprises: generating one or more dynamic link budgets for the flight based on a received flight plan, wherein each dynamic link budget is configured to determine RF spectrum availability of a communications link in one or more coverage areas of the communications network; and determining RF spectrum availability based on the dynamic link budgets corresponding to the one or more coverage areas of the communications network that the aircraft is to fly in based on the received flight plan from the user.

4. The method of claim 3, wherein generating the one or more dynamic link budgets for the flight includes using a dynamic RF coverage prediction tool to populate one or more parameters of the one or more dynamic link budgets.

5. The method of claim 2, wherein selecting the communication resource from the traffic resource pool comprises determining a number of RF spectrum traffic resources to assign to the flight based on a throughput requirement of the flight.

6. The method of claim 5, wherein selecting the communication resource from the traffic resource pool comprises selecting the determined number of RF spectrum traffic resources from the available communication resources in the traffic resource pool.

7. The method of claim 2, wherein the communication resource includes one or more RF spectrum channels, wherein each RF spectrum channel of the one or more RF spectrum channels from the traffic resource pool comprises a plurality of time slots, wherein the plurality of time slots are configured to be allocated to a plurality of aircraft such that the plurality of aircraft each communicate using the RF spectrum channel during their respective time slot.
8. The method of claim 7, wherein determining that the performance of the selected communication resource is above the pre-determined threshold comprises determining whether a selected RF spectrum channel has an available time slot from the plurality of time slots to be allocated to the flight.
9. The method of claim 2, wherein in accordance with determining that the performance of the selected communication resource is below the pre-determined threshold, rejecting the received flight plan and alerting the user that the flight plan has been rejected.
10. The method of claim 2, in accordance with determining that the performance of the selected communication resource is below the pre-determined threshold, modifying a flight plan associated with the received flight plan.
11. The method of claim 10, wherein modifying the received flight plan comprises modifying a location and altitude information of the flight plan based on a class of airspace the flight is being conducted in.
12. The method of claim 10, wherein modifying the received flight plan comprises modifying the location and altitude information of the flight plan based on an RF coverage availability of the communications network.
13. The method of claim 10, wherein modifying the received flight plan comprises modifying the location and altitude information of the flight plan based on RF spectrum traffic resource availability of the communications network.
14. The method of claim 2, in accordance with determining that the performance of the selected communication resource is below the pre-determined threshold, changing or canceling another flight plan in the communications network and alerting the user associated with that flight that their flight plan has been changed or canceled.
15. The method of claim 2, wherein the determination that a performance of the selected communication resource is above the pre-determined threshold is based on a flight plan associated with the flight.
16. The method of claim 15, wherein the determination that a performance of the selected communication resource is above the pre-determined threshold is based on one or more flight plans of flights other than the flight associated with the flight plan.
17. A system for allocating radio frequency (RF) spectrum traffic resources in an air-to-ground communication network of networks, the system comprising: a memory; one or more processors; wherein the memory stores one or more programs that when executed by the one or more processors, cause the one or more processors to: receive operational flight requirements associated with a plurality of flight operations that are to fly in a coverage area of an air-to-ground communications network; determine communication resource availability for a flight that is to fly in the coverage area of the air-to-ground communications network, wherein determining communication resource availability for the flight comprises: receive a traffic resource pool from a communication network, wherein the traffic resource pool includes one or more communication resources; and select a communication resource from the received traffic resource pool, wherein selecting the communication resource from the received traffic resource pool is based on the received operational flight requirements associated with the plurality of flight operations; determine that a performance of the selected communication resource is above a pre-determined threshold, wherein the determination is based on a radio configuration information associated with the flight that is to fly in the coverage area; and in accordance with the determination that the performance of the selected communication resources is above the pre-determined threshold,

operate the flight using the selected communication resource from the received traffic resource pool.

18. A non-transitory computer readable storage medium storing one or more programs for allocating RF spectrum channels in an air-to-ground communications network of networks, for execution by one or more processors of an electronic device that when executed by the device, cause the device to: receive operational flight requirements associated with a plurality of flight operations that are to fly in a coverage area of an air-to-ground communications network; determine communication resource availability for a flight that is to fly in the coverage area of the air-to-ground communications network, wherein determining communication resource availability for the flight comprises: receive a traffic resource pool from a communication network, wherein the traffic resource pool includes one or more communication resources; and select a communication resource from the received traffic resource pool, wherein selecting the communication resource from the received traffic resource pool is based on the received operational flight requirements associated with the plurality of flight operations; determine that a performance of the selected communication resource is above a pre-determined threshold, wherein the determination is based on a radio configuration information associated with the flight that is to fly in the coverage area; and in accordance with the determination that the performance of the selected communication resources is above the pre-determined threshold, operate the flight using the selected communication resource from the received traffic resource pool.
